INTEGRATED ROBUST SOLUTIONS, ISLAMABAD

Week 8 Task: AI Internship

Real-World Machine Learning Systems

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Version: 1.0

Date: August 22, 2025

Prepared for IR Solutions AI Interns

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1 Introduction

This document outlines the Week 8 tasks for AI interns at IR Solutions. The focus of this week is on understanding critical considerations for building and deploying machine learning (ML) models in real-world settings. Interns will study selected guides from Google's Machine Learning resources, covering image classification, rules of ML, people + AI guidebook, text classification, good data analysis, deep learning tuning playbook, data traps, introduction to responsible AI, and adversarial testing for generative AI. To foster creativity and personalized learning, there are no mandatory practical tasks; instead, interns are given the flexibility to explore any topic related to AI, ML, Python, or related technologies. Weekly deliverables include a PDF report, submitted by **Friday, August 29, 2025, 5:30 PM PKT**.

2 Objectives

The Week 8 tasks aim to:

- Develop a comprehensive understanding of real-world ML systems, including image classification, best practices, responsible AI, and advanced tuning techniques.
- Explore automated and manual techniques to streamline model development and ensure fairness and safety.
- Learn strategies for data analysis, avoiding common pitfalls, and adversarial testing in generative AI.
- Encourage independent exploration of AI/ML topics through self-directed projects or theoretical study.
- Practice professional documentation through a structured PDF report summarizing learning and exploration.

3 Theoretical Learning

Interns are required to complete the following guides from Google's Machine Learning resources to gain insights into real-world ML systems:

3.1 Image Classification

- **Course Material**: Image Classification
- **Learning Objectives**:
  - Understand how to build an image classifier to distinguish between categories like cats and dogs.
  - Learn the basics of convolutional neural networks and their application in image classification tasks.

3.2 Rules of ML

- **Course Material**: Rules of ML
- **Learning Objectives**:
  - Become a better machine learning engineer by following machine learning best practices used at Google.
  - Explore guidelines for designing, implementing, and maintaining effective ML systems.

3.3 People + AI Guidebook

- **Course Material**: People + AI Guidebook
- **Learning Objectives**:
  - Assist UXers, PMs, and developers in collaboratively working through AI design topics and questions.
  - Understand human-centered design principles for AI systems.

3.4 Text Classification

- **Course Material**: Text Classification
- **Learning Objectives**:
  - Follow a comprehensive guide to solving text classification problems using machine learning.
  - Learn workflows for topic classification and sentiment analysis.

3.5 Good Data Analysis

- **Course Material**: Good Data Analysis
- **Learning Objectives**:
  - Learn tricks that expert data analysts use to evaluate huge data sets in machine learning problems.
  - Understand technical, process, and mindset aspects of effective data analysis.

3.6 Deep Learning Tuning Playbook

- **Course Material**: Deep Learning Tuning Playbook
- **Learning Objectives**:
  - Explain a scientific way to optimize the training of deep learning models.
  - Cover hyperparameter tuning, training pipelines, and debugging techniques.

3.7 Data Traps

- **Course Material**: Data Traps (Note: Link provided is for Intro to Responsible AI; adjust if needed)
- **Learning Objectives**:
  - Identify common mistakes that ML practitioners might encounter when working with data and statistics.
  - Learn to avoid pitfalls in data handling and analysis.

3.8 Intro to Responsible AI

- **Course Material**: Intro to Responsible AI (Note: Link provided is for Adversarial Testing; adjust if needed)
- **Learning Objectives**:
  - Get an overview of how to build fairness, accountability, safety, and privacy into AI systems.
  - Understand responsible AI dimensions and practices.

3.9 Adversarial Testing for Generative AI

- **Course Material**: https://developers.google.com/machine-learning/guides/adv-testing
- **Learning Objectives**:
  - Learn adversarial testing methods for generative AI models.
  - Understand workflows for identifying and mitigating harms in model outputs.

4 Self-Directed Exploration

For Week 8, there are no mandatory practical tasks. Instead, interns are encouraged to leverage this opportunity to explore any area of interest within artificial intelligence, machine learning, Python, or related technologies. This flexibility allows interns to pursue topics that align with their career goals or curiosity. Possible exploration areas include:

- **Practical Coding**: Develop a project using Python, such as building a machine learning model, experimenting with deep learning frameworks (e.g., TensorFlow, PyTorch), or creating a data visualization dashboard.
- **Model Experimentation**: Revisit and improve models from previous weeks (e.g., breast cancer prediction or Fashion MNIST classification) or explore new datasets and algorithms from platforms like Kaggle or UCI Machine Learning Repository.
- **Theoretical Study**: Dive deeper into AI/ML concepts, such as reinforcement learning, generative AI, computer vision, or natural language processing, through research papers, online courses, or tutorials.
- **Tool Exploration**: Experiment with AI/ML tools and platforms, such as Jupyter notebooks etc.

- **Other AI-Related Topics**: Explore emerging areas like explainable AI, federated learning, or AI ethics, or work on Python-based projects like web scraping, automation scripts, or API development for AI applications.

Interns should document their exploration in the PDF report, including the motivation, methodology, outcomes, and any code, visualizations, or insights produced.

5 Submission Guidelines

5.1 PDF Report

- **Format**:
  - **Title Page**: Include IR Solutions logo, intern's name, designation (AI Intern), department (Artificial Intelligence), and date.
  - **Sections**:
    - **Theoretical Understanding**: Summarize key concepts from the assigned course modules (Image Classification, Rules of ML, People + AI Guidebook, Text Classification, Good Data Analysis, Deep Learning Tuning Playbook, Data Traps, Intro to Responsible AI, Adversarial Testing for Generative AI).
    - **Self-Directed Exploration**: Describe the chosen exploration topic, including motivation, methodology, and outcomes (e.g., code, experiments, or theoretical insights).
    - **Challenges and Approaches**: List challenges faced during exploration and solutions applied.
    - **Results and Analysis**: Present outcomes of the exploration, including any visualizations, metrics, or findings.
    - **Spare Time Activities**: Note any additional learning or activities undertaken during the week.
    - **Conclusion**: Summarize key takeaways and plans for further improvement.
    - **References**: Cite all resources used, including course materials and additional sources.
  - Use professional formatting with headings, subheadings, bullet points, and tables/plots for results (if applicable).
- **File Naming**: `Week8_InternName_Report.pdf`

5.2 Optional Code Submission

- If the self-directed exploration involves coding (e.g., a Google Colab notebook), submit the code as `Week8_InternName_Exploration.ipynb`.
- Follow PEP 8 standards, include clear comments, and use Markdown cells for methodology, analysis, and visualizations (if applicable).
- Ensure the notebook is fully executable with no errors.

5.3 Submission Process

- Upload the PDF report (and optional .ipynb file) to the designated Google Drive folder by **Friday, August 29, 2025, 5:30 PM PKT**.
- Ensure files are clearly named and organized in the Week 8 folder.

6 Evaluation Criteria

Interns will be evaluated based on:

- **Theoretical Understanding (40%)**: Depth and clarity of concepts learned from the course modules.
- **Self-Directed Exploration (30%)**: Creativity, relevance, and quality of the chosen exploration topic and its documentation.
- **Report Quality (20%)**: Clarity, completeness, and adherence to the defined format.
- **Best Practices (10%)**: For code submissions, adherence to PEP 8, modular code, and clear documentation.

7 Additional Notes

- **Resources**: Refer to the Google Machine Learning guides and additional resources like research papers, online tutorials, or AI/ML platforms for self-directed exploration.
- **Support**: Reach out to AI team members or Lead AI via Slack or during the standup meeting (Friday, 3:00 PM PKT) for guidance.
- **Expectations**: Maintain professionalism, meet deadlines, and actively engage in learning and exploration.

8 References

- Google Machine Learning Crash Course: `https://developers.google.com/machine-learn` `crash-course`
- GeeksforGeeks:
  - `https://www.geeksforgeeks.org/machine-learning`
  - `https://www.geeksforgeeks.org/automated-machine-learning-automl`
  - `https://www.geeksforgeeks.org/fairness-in-machine-learning`
- YouTube Tutorials:
  - `https://www.youtube.com/watch?v=_3y7C7_do-k`
  - `https://www.youtube.com/watch?v=7Bg3iBq-SIY`
  - `https://www.youtube.com/watch?v=0YdpwSYMY6I`